ECON 5021 Unit 2: Data Analysis Study Guide

How to type the R for the coding section mechanically. Built from the practice exam 26101200001 (`fertil1 + catholic`). The exam will be a permutation of it: a different `wooldridge` dataset and different variables, with the same 15 question types.

1. The shape of the exam

15 multiple-choice questions in two blocks. Each block starts with *"This is the start of a multipart question"*, names a `wooldridge` dataset, and runs on one model.

Q	Type	One-liner	Main trap
1	Summary stat	`mean(dat$y)`	mean vs median; `na.rm=TRUE`
2	Count NAs	`sum(is.na(dat$x))`	the right variable
3	OLS coefficient	`m = lm(y ~ ..., dat)`	`I(x^2)`; the answer may be (e)
4	SSR	`sum(resid(m)^2)`	SSR vs $\hat{\sigma}$ vs $\hat{\sigma}^2$
5	Critical value	`qt(1-a/2, df.residual(m))`	`qnorm`; df from the full n when NAs were dropped
6	Strongest conclusion	p-value from `summary(m)`	map p to a level (table below)
7	p-value, $H_0 : \beta = c$	$t = (\hat{\beta} - c)/se$	two-sided; $c \neq 0$, so you can't read it off `summary`
8	Cov of two estimates	`vcov(m)["a","b"]`	name is `"(Intercept)"`, `"I(age^2)"`
9	Make a dummy, %	`mean(new)*100`	include every group; NAs
10	LPM R^2	`summary(m1)$r.squared`	R^2 vs adjusted; the NA rows lm drops
11	Interpretation	coef $\times 100$	**percentage points**, not percent
12	CI bound	`confint(m1, level=.90)`	level; which row; lower or upper
13	Strongest conclusion	same as Q6	
14	% change in coef	`100*(new/old-1)`	sign; the ratio alone is not the answer
15	VIF	`car::vif(m2)`	by hand you need the **same rows**

2. Setup: type this first (before reading anything else)

```
install.packages(c("wooldridge", "car"))   # only if not installed (check this BEFORE the exam)
library(wooldridge)
library(car)
dat = wooldridge::fertil1        # <- dataset name from the exam
?fertil1                         # variable descriptions, if a name is unclear
names(dat); nrow(dat)            # spelling + n (you need n to spot dropped rows)
```

3. The master skeleton (memorize this, then fill in the blanks)

Fill in only the parts marked <. . .>. Everything else is the same on every version.

```
# ================ BLOCK 1 (Q1-8) ================
dat = wooldridge::<data1>
mean(dat$<y>)                                  # Q1  (or median / sd / quantile)
sum(is.na(dat$<x>))                            # Q2
m = lm(<y> ~ <x1> + <x2> + I(<x2>^2) + <...>, data = dat)
summary(m)                                     # Q3 coef, Q6 p-value (Pr(>|t|))
nobs(m); df.residual(m)                        # check: were rows dropped?
sum(resid(m)^2)                                # Q4 SSR
qt(1 - <alpha>/2, df = df.residual(m))         # Q5  alpha = 1 - level  (97.5% -> 0.025)
b = coef(m)["<x>"]; se = sqrt(vcov(m)["<x>", "<x>"])
2 * pt(-abs((b - <c>) / se), df = df.residual(m))   # Q7  H0: beta_x = c
vcov(m)["(Intercept)", "<x>"]                  # Q8
```

```
# ================= BLOCK 2 (Q9-15) =================
dat = wooldridge::<data2>
dat$<new> = as.numeric(<condition>)            # e.g. asian == 1 | hispan == 1 | black == 1
100 * mean(dat$<new>, na.rm = TRUE)            # Q9
m1 = lm(<ybin> ~ <new> + <x1> + <x2> + <x3>, data = dat)   # 5 parameters incl. intercept
summary(m1)                                    # Q10 R^2 (bottom), Q13 p-value
100 * coef(m1)["<x>"]                          # Q11 -> "percentage points"
confint(m1, level = <0.90>)                    # Q12
m2 = lm(<ybin> ~ <new> + <x1> + <x2> + <x3> + <extra>, data = dat)
100 * (coef(m2)["<x>"] / coef(m1)["<x>"] - 1)  # Q14
vif(m2)                                        # Q15
```

Speed: `summary(m)` alone answers Q3, Q6, Q10, Q13 and the SE you need for Q7. Run it once and read off all of them. If a name isn't found in `coef(m)["..."]`, print `names(coef(m))` and copy the exact spelling (`"I(age^2)"`, `"(Intercept)"`).

4. Rules you must know cold

Strongest conclusion (Q6, Q13). Take p = `Pr(>|t|)` for that variable:

$p < 0.001$	(a) reject at 0.1% (***)	$0.05 \le p < 0.10$	(d) reject at 10% (.)
$0.001 \le p < 0.01$	(b) reject at 1% (**)	$p \ge 0.10$	(e) fail to reject at 10%
$0.01 \le p < 0.05$	(c) reject at 5% (*)		

The significance stars in `summary()` map one-to-one onto (a)–(d), and no star means (e). `"e-05"` means 10^{-5}, so that's (a).

Critical value (Q5). Confidence level L, so $\alpha = 1 - L$, and the value is `qt(1 - alpha/2, df.residual(m))`.

Level	90%	95%	97.5%	98%	99%	99.5%
`qt(p, df)` with p =	0.95	0.975	0.9875	0.99	0.995	0.9975

df trap. Always use `df.residual(m)`, which equals $n_{\text{used}} - k$. When the model has a variable with NAs, lm silently drops those rows, so $n_{\text{used}} <$ `nrow(dat)`. The wrong-df answer and the `qnorm` answer are both planted as distractors. In practice B the correct value is not even listed, so the answer is (e).

p-value for $H_0 : \beta = c$ (Q7). $t = (\hat{\beta} - c)/se(\hat{\beta})$, then $p = 2\,P(T_{df} > |t|)$ = `2*pt(-abs(t), df)`. The `summary()` p-value tests $c = 0$ only. **Watch the sign of c:** for $H_0 : \beta = -0.15$ the numerator is $\hat{\beta} + 0.15$. The one-sided value (half the answer) is a planted distractor.

Percent vs percentage points (Q11). In an LPM (0/1 outcome) $\hat{\beta} = 0.0302$ means the probability goes up by $0.0302 \times 100 = 3.02$ **percentage points**. "3.02 percent" is wrong.

Model	Interpretation of $\hat{\beta}_x$	Example
$y \in \{0,1\}$ (LPM), x level	$100\hat{\beta}$ percentage points per unit of x	$0.0302 \to 3.02$ pp
$\log y$, x level	$\approx 100\hat{\beta}$ % change in y per unit of x	$0.063 \to 6.3\%$ (exact: $100(e^{\hat{\beta}} - 1)$)
y level, $\log x$	$\hat{\beta}/100$ units of y per 1% increase in x	
$\log y$, $\log x$	elasticity: $\hat{\beta}$ % per 1%	
x and x^2	marginal effect $\hat{\beta}_1 + 2\hat{\beta}_2 x$; turning point $-\hat{\beta}_1/(2\hat{\beta}_2)$	

Percentage change in a coefficient (Q14). $100 \times (\hat{\beta}_{\text{new}}/\hat{\beta}_{\text{old}} - 1)$, where "new" is the model with the extra control. The sign matters. If both coefficients are negative and the new one is closer to zero, the change is negative (e.g. –92%).

VIF (Q15). $\text{VIF}_j = 1/(1 - R_j^2)$, where R_j^2 comes from regressing x_j on all the *other* regressors (not y). The easy way is `car::vif(m2)`, which automatically uses the same rows as `m2`.

Same-observations trap. If the **outcome** has NAs (`hsgrad` in `catholic`, `highwage` in `mroz`), the main regression drops those rows but the auxiliary regression of x_j on the other x's does not, because y isn't in it. By hand you must subset first:

```
d = dat[!is.na(dat$<y>), ]           # or: d = model.frame(m2)
1 / (1 - summary(lm(<xj> ~ <other x's>, data = d))$r.squared)
```

`catholic`: 1.0249 (correct) vs 1.0167 (wrong). `mroz`: 1.5525 vs 1.6279. Other distractors are $1 - R_j^2$ (below 1, impossible for a VIF) and the VIF of a different variable.

Making the new variable (Q9).

```
dat$nonwhite = as.numeric(dat$asian == 1 | dat$hispan == 1 | dat$black == 1)   # OR of groups
dat$young    = as.numeric(dat$age < 35)            # strict vs <= changes the answer
dat$arrested = as.numeric(dat$narr86 > 0)          # count -> any
dat$highwage = as.numeric(dat$wage > 5)            # NA stays NA -> need na.rm = TRUE
100 * mean(dat$young)                              # "what percent" -> times 100
```

NA checklist. (1) `mean()` of a column with NAs returns `NA`, so add `na.rm = TRUE`. (2) Read the denominator off the question: "percent of the population" versus "among those with a reported wage". (3) After `lm`, compare `nobs(m)` with `nrow(dat)`. If they differ, every df-based answer (Q5, Q7, Q12) has to use `df.residual(m)`, and `confint`/`summary` already do.

5. Likely twists (same muscle, a different function)

```
median(x); sd(x); var(x); IQR(x)
quantile(x, 0.9); min(x); max(x)
table(dat$x); mean(dat$x == 2)
sum(complete.cases(dat))       # rows w/o NA
summary(m)$adj.r.squared
summary(m)$sigma               # resid. SE
summary(m)$sigma^2             # SSR/df
summary(m)$fstatistic          # overall F
nobs(m); df.residual(m)
coef(summary(m))["x", ]        # est,se,t,p
sqrt(diag(vcov(m)))            # all SEs
```

```
pt(t, df, lower.tail = FALSE)  # 1-sided
# joint test H0: b1 = b2 = 0
linearHypothesis(m, c("x1 = 0", "x2 = 0"))
anova(m_restricted, m)         # same F
# prediction
predict(m, newdata = data.frame(educ = 12,
  age = 40, year = 80, ...))
# CI by hand
b + c(-1, 1) * qt(.95, df) * se
# turning point of a quadratic
-coef(m)["age"] / (2 * coef(m)["I(age^2)"])
```

6. Execution checklist (about 30 seconds per question)

1. Type the setup and `dat = wooldridge::....` Run `names(dat)` and `nrow(dat)`.
2. Q1–Q2: one line each. Q2 just counts; it doesn't change the data.
3. Write the model **exactly** as printed. Squares are `I(age^2)` (or an existing `agesq` column; both give identical estimates). Run `summary(m)` once.
4. Read Q3, Q6 and the SE for Q7 from that one output. Q4 `sum(resid(m)^2)`, Q5 `qt`, Q8 `vcov`.
5. Block 2: create the dummy **before** fitting. Count the parameters (the question says "5 parameters including the intercept").
6. Q11: multiply by 100 and say *percentage points*. Q12: `confint` at the stated level.
7. Q14–Q15: fit `m2` = `m1` + the extra control. Use `vif(m2)`.
8. None of the above is a real answer, used about 2–3 times per exam. If your number is computed correctly and isn't listed, pick (e). Don't round toward the nearest option.

7. Practice set in this folder

File	Block 1	Block 2	Twists
`Original_Key / _Highlighted`	`fertil1`	`catholic`	the real practice exam
`PracticeA_Key`	`bwght` (NAs)	`k401ksubs`	99% t, $c < 0$ null, upper CI bound
`PracticeB_Key`	`wage2` (log y)	`crime1`	median, $\hat{\sigma}$, 98% t not listed, adj. R^2
`PracticeC_Key`	`card`	`mroz` (NA outcome)	`na.rm`, `nobs`, VIF same-rows trap

How to drill: open a blank R script, type the Section 3 skeleton from memory, then work an exam with the key hidden. Check against the `key_*.r` file. Target: all 15 questions in under 20 minutes.